



Configuring Authentication

Authentication provides a method to identify users, which includes the login and password dialog, challenge and response, messaging support, and encryption, depending on the selected security protocol. Authentication is the way a user is identified prior to being allowed access to the network and network services.

- [Prerequisites for Configuring Authentication, on page 1](#)
- [Restrictions for Configuring Authentication, on page 1](#)
- [Information About Authentication, on page 2](#)
- [How to Configure Authentication, on page 17](#)
- [Configuration Examples for Authentication, on page 34](#)
- [Feature History for Configuring Authentication, on page 47](#)

Prerequisites for Configuring Authentication

The implementation of authentication is divided into Authentication, Authorization, and Accounting (AAA) authentication and nonauthentication methods. Cisco recommends that, whenever possible, AAA security services be used to implement authentication.

Restrictions for Configuring Authentication

- The number of AAA method lists that can be configured is 250.
- If you configure the same RADIUS server IP address for a different UDP destination port for accounting requests by using the **acct-port** keyword and a UDP destination port for authentication requests by using the **auth-port** keyword with and without the nonstandard option, the RADIUS server does not accept the nonstandard option.
- The **aaa accounting delay-start** command is not supported for binOS use cases like dot1x authentication.

Information About Authentication

Named Method Lists for Authentication

A named list of authentication methods is first defined before AAA authentication can be configured, and the named list is then applied to various interfaces. The method list defines the types of authentication and the sequence in which they are performed; it must be applied to a specific interface before any of the defined authentication methods are performed. The only exception is the default method list (which is named “default”). The default method list is automatically applied to all interfaces, except those that have a named method list explicitly defined. A defined method list overrides the default method list.

A method list is a sequential list describing the authentication methods to be queried to authenticate a user. Method lists enable you to designate one or more security protocols to be used for authentication, thus ensuring a backup system for authentication in case the initial method fails. Cisco software uses the first listed method to authenticate users. If that method fails to respond, the Cisco software selects the next authentication method listed in the method list. This process continues until there is successful communication with a listed authentication method, or all methods defined in the method list are exhausted.

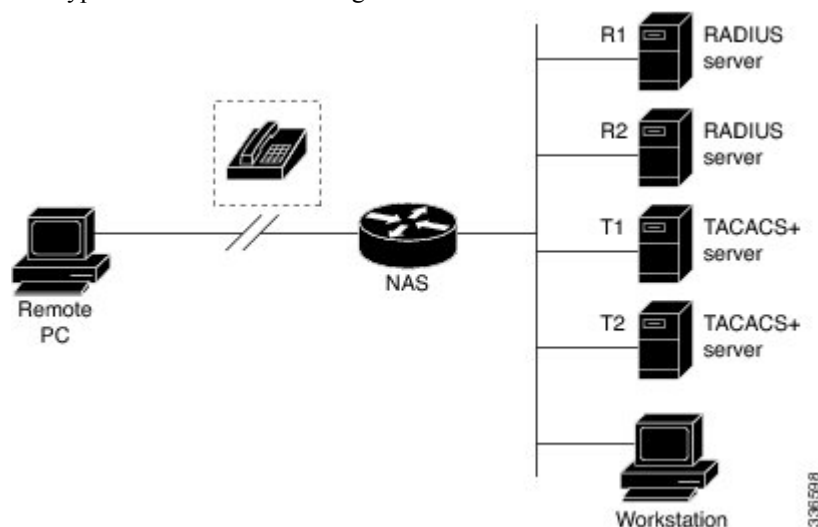
Note that the software attempts authentication with the next listed authentication method only when there is no response from the previous method. If authentication fails at any point in this cycle, that is, the security server or local username database responds by denying the user access, then the authentication process stops and no other authentication methods are attempted.

Method Lists and Server Groups

A server group is a way to group existing RADIUS or TACACS+ server hosts for use in method lists. The figure below shows a typical AAA network configuration that includes four security servers: R1 and R2 are RADIUS servers and T1 and T2 are TACACS+ servers. R1 and R2 make up the group of RADIUS servers. T1 and T2 make up the group of TACACS+ servers.

Figure 1: Configuration of a Typical AAA Network

This figure shows a typical AAA network configuration.



Using server groups, you can specify a subset of the configured server hosts and use them for a particular service. For example, server groups allow you to define R1 and R2 as a server group, and define T1 and T2 as a separate server group. For example, you can specify R1 and T1 in the method list for authentication login, while specifying R2 and T2 in the method list for PPP authentication.

Server groups also can include multiple host entries for the same server, as long as each entry has a unique identifier. The combination of an IP address and a UDP port number creates a unique identifier, allowing different ports to be individually defined as RADIUS hosts providing a specific AAA service. In other words, this unique identifier enables RADIUS requests to be sent to different UDP ports on a server at the same IP address. If two different host entries on the same RADIUS server are configured for the same service—for example, authentication--the second host entry configured acts as failover backup to the first one. Using this example, if the first host entry fails to provide accounting services, the network access server will try the second host entry configured on the same device for accounting services. (The RADIUS host entries will be tried in the order in which they are configured.)

For more information about configuring server groups and about configuring server groups based on Dialed Number Identification Service (DNIS) numbers, see the “Configuring RADIUS” or “Configuring TACACS+” chapters.

Login Authentication Using AAA

Login Authentication Using Enable Password

Use the **aaa authentication login** command with the **enable** keyword to specify the enable password as the login authentication method. For example, to specify the enable password as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication login default enable
```

Before you can use the enable password as the login authentication method, you need to define the enable password. For more information about defining enable passwords, see chapter “Controlling Switch Access with Passwords and Privilege Levels.”

Login Authentication Using Line Password

Use the **aaa authentication login default** command with the **line** keyword to specify the line password as the login authentication method. For example, to specify the line password as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication login default line
```

Before you can use a line password as the login authentication method, you need to define a line password.

Login Authentication Using Local Password

Use the **aaa authentication login default** command with the **local** keyword to specify that the Cisco device will use the local username database for authentication. For example, to specify the local username database as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication login default local
```

Login Authentication Using Group RADIUS

Use the **aaa authentication login default** command with the **group radius** to specify RADIUS as the login authentication method. For example, to specify RADIUS as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication login default group radius
```

Before you can use RADIUS as the login authentication method, you need to enable communication with the RADIUS security server. For more information about establishing communication with a RADIUS server, refer to the chapter “Configuring RADIUS.”

RADIUS Attribute 8 in Access Requests

After you have used the **aaa authentication login** command to specify RADIUS and your login host has been configured to request its IP address from the NAS, you can send attribute 8 (Framed-IP-Address) in access-request packets by using the **radius-server attribute 8 include-in-access-req** command in global configuration mode. This command makes it possible for NAS to provide the RADIUS server a hint of the user IP address in advance for user authentication.

Login Authentication Using Group TACACS

Use the **aaa authentication login default** command with the **group tacacs+** to specify TACACS+ as the login authentication method. For example, to specify TACACS+ as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication login default group tacacs+
```

Before you can use TACACS+ as the login authentication method, you need to enable communication with the TACACS+ security server. For more information about establishing communication with a TACACS+ server, refer to the chapter “Configuring TACACS+.”



Note If the **ip radius source interface** command is configured under the group, the MTU of the source interface is used for injecting the packet. And if not configured, default fragmentation at 1396 bytes is done.

Login Authentication Using Group Name

Use the **aaa authentication login default** command with the **group group-name** method to specify a subset of RADIUS or TACACS+ servers to use as the login authentication method. To specify and define the group name and the members of the group, use the **aaa group server** command. For example, use the **aaa group server** command to first define the members of **group loginrad**:

```
Device> enable
Device# configure terminal
Device(config)# aaa group server radius loginrad
Device(config-sg-radius)# server 172.16.2.3
Device(config-sg-radius)# server 172.16.2.17
Device(config-sg-radius)# server 172.16.2.32
Device(config-sg-radius)# end
```

This command specifies RADIUS servers 172.16.2.3, 172.16.2.17, and 172.16.2.32 as members of the group *loginrad*.

To specify **group loginrad** as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication login default group loginrad
```

Before you can use a group name as the login authentication method, you need to enable communication with the RADIUS or TACACS+ security server. For more information about establishing communication with a RADIUS server, refer to the chapter “Configuring RADIUS.” For more information about establishing communication with a TACACS+ server, refer to the chapter “Configuring TACACS+.”

PPP Authentication Using AAA

PPP Authentication Using Local Password

Use the **aaa authentication ppp default** command with the **local** keyword to specify that the Cisco device will use the local username database for authentication. For example, to specify the local username database as the method of authentication for use on lines running PPP when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication ppp default local
```

PPP Authentication Using Group RADIUS

Use the **aaa authentication ppp default group radius** command to specify RADIUS as the login authentication method. For example, to specify RADIUS as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication ppp default group radius
```

Before you can use RADIUS as the PPP authentication method, you need to enable communication with the RADIUS security server. For more information about establishing communication with a RADIUS server, refer to the chapter “Configuring RADIUS.”

RADIUS Attribute 44 in Access Requests

After you have used the **aaa authentication ppp default group radius** command to specify RADIUS as the login authentication method, you can configure your device to send attribute 44 (Acct-Session-ID) in access-request packets by using the **radius-server attribute 44 include-in-access-req** command in global configuration mode. This command allows the RADIUS daemon to track a call from the beginning to the end.

PPP Authentication Using Group TACACS

Use the **aaa authentication ppp default group tacacs+** command to specify TACACS+ as the login authentication method. For example, to specify TACACS+ as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication ppp default group tacacs+
```

Before you can use TACACS+ as the PPP authentication method, you need to enable communication with the TACACS+ security server. For more information about establishing communication with a TACACS+ server, refer to the chapter “Configuring TACACS+.”

PPP Authentication Using Group Name

Use the **aaa authentication ppp default** command with the **group group-name** method to specify a subset of RADIUS or TACACS+ servers to use as the login authentication method. To specify and define the group name and the members of the group, use the **aaa group server** command. For example, use the **aaa group server** command to first define the members of **group ppprad**:

```

Device> enable
Device# configure terminal
Device(config)# aaa group server radius ppprad
Device(config-sg-radius)# server 172.16.2.3
Device(config-sg-radius)# server 172.16.2.17
Device(config-sg-radius)# server 172.16.2.32
Device(config-sg-radius)# end

```

This command specifies RADIUS servers 172.16.2.3, 172.16.2.17, and 172.16.2.32 as members of the group *ppprad*.

To specify **group ppprad** as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication ppp default group ppprad
```

Before you can use a group name as the PPP authentication method, you need to enable communication with the RADIUS or TACACS+ security server. For more information about establishing communication with a RADIUS server, refer to the chapter “Configuring RADIUS”. For more information about establishing communication with a TACACS+ server, refer to the chapter “Configuring TACACS+.”

AAA Scalability for PPP Requests

You can configure and monitor the number of background processes allocated by the PPP manager in the network access server (NAS) to deal with AAA authentication and authorization requests. The AAA Scalability feature enables you to configure the number of processes used to handle AAA requests for PPP, thus increasing the number of users that can be simultaneously authenticated or authorized.

To allocate a specific number of background processes to handle AAA requests for PPP, use the following command in global configuration mode:

```
Device(config)# aaa processes 5000
```

The argument *number* defines the number of background processes earmarked to process AAA authentication and authorization requests for PPP and can be configured for any value from 1 to 2147483647. Because of the way the PPP manager handles requests for PPP, this argument also defines the number of new users that can be simultaneously authenticated. This argument can be increased or decreased at any time.



Note Allocating additional background processes can be expensive. You should configure the minimum number of background processes capable of handling the AAA requests for PPP.

ARAP Authentication Using AAA

ARAP Authentication Allowing Authorized Guest Logins

Use the **aaa authentication arap default** command with the **auth-guest** keyword to allow guest logins only if the user has already successfully logged in to the EXEC. This method must be the first listed in the ARAP authentication method list but it can be followed by other methods if it does not succeed. For example, to allow all authorized guest logins--meaning logins by users who have already successfully logged in to the EXEC--as the default method of authentication, using RADIUS only if that method fails, enter the following command:

```
Device(config)# aaa authentication arap default auth-guest group radius
```



Note By default, guest logins through ARAP are disabled when you initialize AAA. To allow guest logins, you must use the **aaa authentication arap** {*authentication-list* | **default** command with either the **guest** or the **auth-guest** keyword.

ARAP Authentication Allowing Guest Logins

Use the **aaa authentication arap** {**default** | *authentication-list*} command with the **guest** keyword to allow guest logins. This method must be the first listed in the ARAP authentication method list but it can be followed by other methods if it does not succeed. For example, to allow all guest logins as the default method of authentication, using RADIUS only if that method fails, enter the following command:

```
Device(config)# aaa authentication arap default guest group radius
```

ARAP Authentication Using Line Password

Use the **aaa authentication arap** {**default** | *authentication-list*} command with the **line** keyword to specify the line password as the authentication method. For example, to specify the line password as the method of ARAP user authentication when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication arap default line
```

Before you can use a line password as the ARAP authentication method, you need to define a line password.

ARAP Authentication Using Local Password

Use the **aaa authentication arap** {**default** | *authentication-list*} command with the **local** keyword to specify that the Cisco device will use the local username database for authentication. For example, to specify the local username database as the method of ARAP user authentication when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication arap default local
```

ARAP Authentication Using Group RADIUS

Use the **aaa authentication arap** {**default** | *authentication-list*} command with the **group radius** method to specify RADIUS as the ARAP authentication method. For example, to specify RADIUS as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication arap default group radius
```

Before you can use RADIUS as the ARAP authentication method, you need to enable communication with the RADIUS security server. For more information about establishing communication with a RADIUS server, refer to the chapter “Configuring RADIUS.”

ARAP Authentication Using Group TACACS

Use the **aaa authentication arap** {**default** | *authentication-list*} command with the **group tacacs+ method** to specify TACACS+ as the ARAP authentication method. For example, to specify TACACS+ as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication arap default group tacacs+
```

Before you can use TACACS+ as the ARAP authentication method, you need to enable communication with the TACACS+ security server. For more information about establishing communication with a TACACS+ server, refer to the chapter “Configuring TACACS+.”

ARAP Authentication Using a Group Name

Use the **aaa authentication arap** {**default** | *authentication-list*} command with the **group group-name** method to specify a subset of RADIUS or TACACS+ servers to use as the ARAP authentication method. To specify and define the group name and the members of the group, use the **aaa group server** command. For example, use the **aaa group server** command to first define the members of **group araprad**:

```
Device> enable
Device# configure terminal
Device(config)# aaa group server radius araprad
Device(config-sg-radius)# server 172.16.2.3
Device(config-sg-radius)# server 172.16.2.17
Device(config-sg-radius)# server 172.16.2.32
Device(config-sg-radius)# end
```

This command specifies RADIUS servers 172.16.2.3, 172.16.2.17, and 172.16.2.32 as members of the group *araprad*.

To specify **group araprad** as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication arap default group araprad
```

Before you can use a group name as the ARAP authentication method, you need to enable communication with the RADIUS or TACACS+ security server. For more information about establishing communication with a RADIUS server, refer to the chapter “Configuring RADIUS.” For more information about establishing communication with a TACACS+ server, refer to the chapter “Configuring TACACS+.”

NASl Authentication Using AAA

NASl Authentication Using Enable Password

Use the **aaa authentication nasi** command with the keyword **enable** to specify the enable password as the authentication method. For example, to specify the enable password as the method of NASl user authentication when no other method list has been defined, use the following command:

```
Device(config)# aaa authentication nasi default enable
```

Before you can use the enable password as the authentication method, you need to define the enable password.

NASl Authentication Using Group RADIUS

Use the **aaa authentication nasi** command with the **group radius** method to specify RADIUS as the NASl authentication method. For example, to specify RADIUS as the method of NASl user authentication when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication nasi default group radius
```

Before you can use RADIUS as the NASl authentication method, you need to enable communication with the RADIUS security server.

NASI Authentication Using Group TACACS

Use the **aaa authentication nasi** command with the **group tacacs+** keyword to specify TACACS+ as the NASI authentication method. For example, to specify TACACS+ as the method of NASI user authentication when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication nasi default group tacacs+
```

Before you can use TACACS+ as the authentication method, you need to enable communication with the TACACS+ security server.

NASI Authentication Using Line Password

Use the **aaa authentication nasi** command with the keyword **line** to specify the line password as the authentication method. For example, to specify the line password as the method of NASI user authentication when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication nasi default line
```

Before you can use a line password as the NASI authentication method, you need to define a line password.

NASI Authentication Using Local Password

Use the **aaa authentication nasi** command with the keyword **local** to specify that the Cisco rdevice will use the local username database for authentication information. For example, to specify the local username database as the method of NASI user authentication when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication nasi default local
```

NASI Authentication Using Group Name

Use the **aaa authentication nasi** command with the **group group-name** method to specify a subset of RADIUS or TACACS+ servers to use as the NASI authentication method. To specify and define the group name and the members of the group, use the **aaa group server** command. For example, use the **aaa group server** command to first define the members of **group nasirad**:

```
Device> enable
Device# configure terminal
Device(config)# aaa group server radius nasirad
Device(config-sg-radius)# server 172.16.2.3
Device(config-sg-radius)# server 172.16.2.17
Device(config-sg-radius)# server 172.16.2.32
Device(config-sg-radius)# end
```

This command specifies RADIUS servers 172.16.2.3, 172.16.2.17, and 172.16.2.32 as members of the group *nasirad*.

To specify **group nasirad** as the method of user authentication at login when no other method list has been defined, enter the following command:

```
Device(config)# aaa authentication nasi default group nasirad
```

Before you can use a group name as the NASI authentication method, you need to enable communication with the RADIUS or TACACS+ security server.

Specifying the Amount of Time for Login Input

The **timeout login response** command allows you to specify how long the system will wait for login input (such as username and password) before timing out. The default login value is 30 seconds; with the **timeout login response** command, you can specify a timeout value from 1 to 300 seconds. To change the login timeout value from the default of 30 seconds, use the following command in line configuration mode:

```
Device(config-line)# timeout login response 30
```

Password Protection at the Privileged Level

Use the **aaa authentication enable default** command to create a series of authentication methods that are used to determine whether a user can access the privileged EXEC command level. You can specify up to four authentication methods. The additional methods of authentication are used only if the previous method returns an error, not if it fails. To specify that the authentication should succeed even if all methods return an error, specify **none** as the final method in the command line.

Use the following command in global configuration mode:

```
Device(config)# authentication enable default radius
```

or

```
Device(config)# authentication enable default tacacs
```

Changing the Text Displayed at the Password Prompt

Use the **aaa authentication password-prompt** command to change the default text that the Cisco IOS XE software displays when prompting a user to enter a password. This command changes the password prompt for the enable password as well as for login passwords that are not supplied by remote security servers. The **no** form of this command returns the password prompt to the following default value:

```
Password:
```

The **aaa authentication password-prompt** command does not change any dialog that is supplied by a remote TACACS+ or RADIUS server.

The **aaa authentication password-prompt** command works when RADIUS is used as the login method. You will be able to see the password prompt defined in the command shown even when the RADIUS server is unreachable. The **aaa authentication password-prompt** command does not work with TACACS+. TACACS+ supplies the NAS with the password prompt to display to the users. If the TACACS+ server is reachable, the NAS gets the password prompt from the server and uses that prompt instead of the one defined in the **aaa authentication password-prompt** command. If the TACACS+ server is not reachable, the password prompt defined in the **aaa authentication password-prompt** command may be used.

Use the following command in global configuration mode:

```
Device(config)# aaa authentication password-prompt "Enter your password now:"
```

Double Authentication of PPP Sessions

PPP sessions can be authenticated only by using a single authentication method: either PAP or CHAP. Double authentication requires remote users to pass a second stage of authentication (after CHAP or PAP authentication) before gaining network access.

This second (“double”) authentication requires a password that is known to the user but *not* stored on the user’s remote host. Therefore, the second authentication is specific to a user, not to a host. This provides an additional level of security that will be effective even if information from the remote host is stolen. In addition, this also provides greater flexibility by allowing customized network privileges for each user.

The second stage authentication can use one-time passwords such as token card passwords, which are not supported by CHAP. If one-time passwords are used, a stolen user password is of no use to the perpetrator.

How Double Authentication Works

With double authentication, there are two authentication/authorization stages. These two stages occur after a remote user dials in and a PPP session is initiated.

In the first stage, the user logs in using the remote host name; CHAP (or PAP) authenticates the remote host, and then PPP negotiates with AAA to authorize the remote host. In this process, the network access privileges associated with the remote host are assigned to the user.



Note We suggest that the network administrator restrict authorization at this first stage to allow only Telnet connections to the local host.

In the second stage, the remote user must Telnet to the network access server to be authenticated. When the remote user logs in, the user must be authenticated with AAA login authentication. The user then must enter the **access-profile** command to be reauthorized using AAA. When this authorization is complete, the user has been double authenticated, and can access the network according to per-user network privileges.

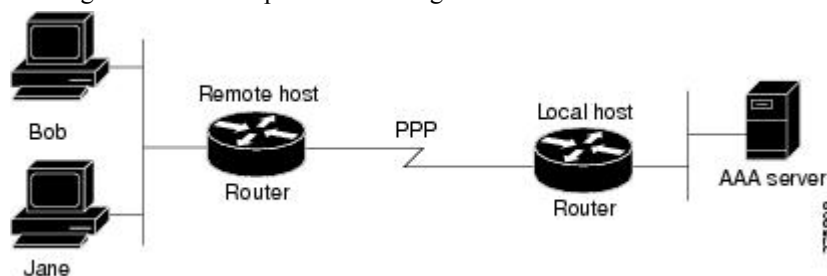
The system administrator determines what network privileges remote users will have after each stage of authentication by configuring appropriate parameters on a security server. To use double authentication, the user must activate it by issuing the **access-profile** command.



Caution Double authentication can cause certain undesirable events if multiple hosts share a PPP connection to a network access server, as shown in the figure below. First, if a user, Bob, initiates a PPP session and activates double authentication at the network access server (per the figure below), any other user will automatically have the same network privileges as Bob until Bob’s PPP session expires. This happens because Bob’s authorization profile is applied to the network access server’s interface during the PPP session and any PPP traffic from other users will use the PPP session Bob established. Second, if Bob initiates a PPP session and activates double authentication, and then--before Bob’s PPP session has expired--another user, Jane, executes the **access-profile** command (or, if Jane Telnets to the network access server and **auto** command **access-profile** is executed), a reauthorization will occur and Jane’s authorization profile will be applied to the interface--replacing Bob’s profile. This can disrupt or halt Bob’s PPP traffic, or grant Bob additional authorization privileges Bob should not have.

Figure 2: Possibly Risky Topology: Multiple Hosts Sharing a PPP Connection to a Network Access Server

This figure shows multiple hosts sharing a PPP connection to a network access server.



Accessing the User Profile After Double Authentication

In double authentication, when a remote user establishes a PPP link to the local host using the local host name, the remote host is CHAP (or PAP) authenticated. After CHAP (or PAP) authentication, PPP negotiates with AAA to assign network access privileges associated with the remote host to the user. (We suggest that privileges at this stage be restricted to allow the user to connect to the local host only by establishing a Telnet connection.)

When the user needs to initiate the second phase of double authentication, establishing a Telnet connection to the local host, the user enters a personal username and password (different from the CHAP or PAP username and password). This action causes AAA reauthentication to occur according to the personal username/password. The initial rights associated with the local host, though, are still in place. By using the **access-profile** command, the rights associated with the local host are replaced by or merged with those defined for the user in the user's profile.

To access the user profile after double authentication, use the following command in EXEC configuration mode:

```
Device> access-profile merge ignore-sanity-checks
```

If you configured the **access-profile** command to be executed as an autocommand, it will be executed automatically after the remote user logs in.

CHAP or PAP Authentication

One of the most common transport protocols used in ISPs dial solutions is the Point-to-Point Protocol (PPP). Traditionally, remote users dial in to an access server to initiate a PPP session. After PPP has been negotiated, remote users are connected to the ISP network and to the Internet.

Because ISPs want only customers to connect to their access servers, remote users are required to authenticate to the access server before they can start up a PPP session. Normally, a remote user authenticates by typing in a username and password when prompted by the access server. Although this is a workable solution, it is difficult to administer and awkward for the remote user.

A better solution is to use the authentication protocols built into PPP. In this case, the remote user dials in to the access server and starts up a minimal subset of PPP with the access server. This does not give the remote user access to the ISP's network--it merely allows the access server to talk to the remote device.

PPP currently supports two authentication protocols: Password Authentication Protocol (PAP) and Challenge Handshake Authentication Protocol (CHAP). Both are specified in RFC 1334 and are supported on synchronous and asynchronous interfaces. Authentication via PAP or CHAP is equivalent to typing in a username and password when prompted by the server. CHAP is considered to be more secure because the remote user's password is never sent across the connection.

PPP (with or without PAP or CHAP authentication) is also supported in dialout solutions. An access server utilizes a dialout feature when it initiates a call to a remote device and attempts to start up a transport protocol such as PPP.



Note To use CHAP or PAP, you must be running PPP encapsulation.

When CHAP is enabled on an interface and a remote device attempts to connect to it, the access server sends a CHAP packet to the remote device. The CHAP packet requests or “challenges” the remote device to respond. The challenge packet consists of an ID, a random number, and the host name of the local device.

When the remote device receives the challenge packet, it concatenates the ID, the remote device’s password, and the random number, and then encrypts all of it using the remote device’s password. The remote device sends the results back to the access server, along with the name associated with the password used in the encryption process.

When the access server receives the response, it uses the name it received to retrieve a password stored in its user database. The retrieved password should be the same password the remote device used in its encryption process. The access server then encrypts the concatenated information with the newly retrieved password--if the result matches the result sent in the response packet, authentication succeeds.

The benefit of using CHAP authentication is that the remote device’s password is never transmitted in clear text. This prevents other devices from stealing it and gaining illegal access to the ISP’s network.

CHAP transactions occur only at the time a link is established. The access server does not request a password during the rest of the call. (The local device can, however, respond to such requests from other devices during a call.)

When PAP is enabled, the remote device attempting to connect to the access server is required to send an authentication request. If the username and password specified in the authentication request are accepted, the Cisco IOS XE software sends an authentication acknowledgment.

After you have enabled CHAP or PAP, the access server will require authentication from remote devices dialing in to the access server. If the remote device does not support the enabled protocol, the call will be dropped.

To use CHAP or PAP, you must perform the following tasks:

- Enable PPP encapsulation.
- Enable CHAP or PAP on the interface.
- For CHAP, configure host name authentication and the secret or password for each remote system with which authentication is required.

Enabling PPP Encapsulation

To enable PPP encapsulation, use the following command in interface configuration mode:

```
Device(config-if) # encapsulation ppp
```

This command enables PPP on an interface.

Enabling PAP or CHAP

To enable CHAP or PAP authentication on an interface configured for PPP encapsulation, use the following command in interface configuration mode:

```
Device(config-if)# ppp authentication chap pap
```

Defines the authentication protocols supported and the order in which they are used. In this command, *protocol1*, *protocol2* represent the following protocols: CHAP, MS-CHAP, and PAP. PPP authentication is attempted first using the first authentication method, which is *protocol1*. If *protocol1* is unable to establish authentication, the next configured protocol is used to negotiate authentication.

If you configure **ppp authentication chap** on an interface, all incoming calls on that interface that initiate a PPP connection will have to be authenticated using CHAP; likewise, if you configure **ppp authentication pap**, all incoming calls that start a PPP connection will have to be authenticated via PAP. If you configure **ppp authentication chap pap**, the access server will attempt to authenticate all incoming calls that start a PPP session with CHAP. If the remote device does not support CHAP, the access server will try to authenticate the call using PAP. If the remote device does not support either CHAP or PAP, authentication will fail and the call will be dropped. If you configure **ppp authentication pap chap**, the access server will attempt to authenticate all incoming calls that start a PPP session with PAP. If the remote device does not support PAP, the access server will try to authenticate the call using CHAP. If the remote device does not support either protocol, authentication will fail and the call will be dropped. If you configure the **ppp authentication** command with the **callin** keyword, the access server will only authenticate the remote device if the remote device initiated the call.

Authentication method lists and the **one-time** keyword are only available if you have enabled AAA; these will not be available if you are using TACACS or extended TACACS. If you specify the name of an authentication method list with the **ppp authentication** command, PPP will attempt to authenticate the connection using the methods defined in the specified method list. If AAA is enabled and no method list is defined by name, PPP will attempt to authenticate the connection using the methods defined as the default. The **ppp authentication** command with the **one-time** keyword enables support for one-time passwords during authentication.

The **if-needed** keyword is only available if you are using TACACS or extended TACACS. The **ppp authentication** command with the **if-needed** keyword means that PPP will only authenticate the remote device via PAP or CHAP if they have not yet authenticated during the life of the current call. If the remote device authenticated via a standard login procedure and initiated PPP from the EXEC prompt, PPP will not authenticate via CHAP if **ppp authentication chap if-needed** is configured on the interface.



Caution

If you use a *list-name* that has not been configured with the **aaa authentication ppp** command, you disable PPP on the line.

Inbound and Outbound Authentication

PPP supports two-way authentication. Normally, when a remote device dials in to an access server, the access server requests that the remote device prove that it is allowed access. This is known as inbound authentication. At the same time, the remote device can also request that the access server prove that it is who it says it is. This is known as outbound authentication. An access server also does outbound authentication when it initiates a call to a remote device.

Enabling Outbound PAP Authentication

To enable outbound PAP authentication, use the following command in interface configuration mode:

```
Device(config-if) # ppp pap sent-username username1 password password1
```

The access server uses the username and password specified by the **ppp pap sent-username** command to authenticate itself whenever it initiates a call to a remote device or when it has to respond to a remote device's request for outbound authentication.

Refusing PAP Authentication Requests

To refuse PAP authentication from peers requesting it, meaning that PAP authentication is disabled for all calls, use the following command in interface configuration mode:

```
Device(config-if) # ppp pap refuse
```

If the **refuse** keyword is not used, the device will not refuse any PAP authentication challenges received from the peer.

Creating a Common CHAP Password

For remote CHAP authentication, you can configure your device to create a common CHAP secret password to use in response to challenges from an unknown peer. For example, if your device calls a rotary of devices (either from another vendor, or running an older version of the Cisco IOS XE software) to which a new (that is, unknown) device has been added. The **ppp chap password** command allows you to replace several username and password configuration commands with a single copy of this command on any dialer interface or asynchronous group interface.

To enable a device calling a collection of devices to configure a common CHAP secret password, use the following command in interface configuration mode:

```
Device(config-if) # ppp chap password secret
```

Refusing CHAP Authentication Requests

To refuse CHAP authentication from peers requesting it, meaning that CHAP authentication is disabled for all calls, use the following command in interface configuration mode:

```
Device(config-if) # ppp chap refuse calling
```

If the **calling** keyword is used, the device will refuse to answer CHAP authentication challenges received from the peer, but will still require the peer to answer any CHAP challenges the device sends.

If outbound PAP has been enabled (using the **ppp pap sent-username** command), PAP is used as the authentication method in the refusal packet.

Delaying CHAP Authentication Until Peer Authenticates

To specify that the device will not authenticate to a peer requesting CHAP authentication until after the peer has authenticated itself to the device, use the following command in interface configuration mode:

```
Device(config-if) # ppp chap wait secret
```

This command (which is the default) specifies that the device will not authenticate to a peer requesting CHAP authentication until the peer has authenticated itself to the device. The **no ppp chap wait** command specifies that the device will respond immediately to an authentication challenge.

Using MS-CHAP

Microsoft Challenge Handshake Authentication Protocol (MS-CHAP) is the Microsoft version of CHAP and is an extension of RFC 1994. Like the standard version of CHAP, MS-CHAP is used for PPP authentication; in this case, authentication occurs between a PC using Microsoft Windows NT or Microsoft Windows 95 and a Cisco device or access server acting as a network access server.

MS-CHAP differs from the standard CHAP as follows:

- MS-CHAP is enabled by negotiating CHAP Algorithm 0x80 in LCP option 3, Authentication Protocol.
- The MS-CHAP Response packet is in a format designed to be compatible with Microsoft Windows NT 3.5 and 3.51, Microsoft Windows 95, and Microsoft LAN Manager 2.x. This format does not require the authenticator to store a clear or reversibly encrypted password.
- MS-CHAP provides an authenticator-controlled authentication retry mechanism.
- MS-CHAP provides an authenticator-controlled change password mechanism.
- MS-CHAP defines a set of “reason-for failure” codes returned in the Failure packet message field.

Depending on the security protocols you have implemented, PPP authentication using MS-CHAP can be used with or without AAA security services. If you have enabled AAA, PPP authentication using MS-CHAP can be used in conjunction with both TACACS+ and RADIUS. The table below lists the vendor-specific RADIUS attributes (IETF Attribute 26) that enable RADIUS to support MS-CHAP.

Table 1: Vendor-Specific RADIUS Attributes for MS-CHAP

Vendor-ID Number	Vendor-Type Number	Vendor-Proprietary Attribute	Description
311	11	MSCHAP-Challenge	Contains the challenge sent by a network access server to an MS-CHAP user. It can be used in both Access-Request and Access-Challenge packets.
211	11	MSCHAP-Response	Contains the response value provided by a PPP MS-CHAP user in response to the challenge. It is only used in Access-Request packets. This attribute is identical to the PPP CHAP Identifier.

Domain Stripping

The AAA Broadcast Accounting feature allows accounting information to be sent to multiple AAA servers at the same time, that is, accounting information can be broadcast to one or more AAA servers simultaneously. This functionality allows you to send accounting information to private and public AAA servers. It also provides redundant billing information for voice applications.

The Domain Stripping feature allows domain stripping to be configured at the server group level.

Per-server group configuration overrides the global configuration. If domain stripping is not enabled globally, but it is enabled in a server group, then it is enabled only for that server group. Also, if virtual routing and forwarding (VRF)-specific domain stripping is configured globally and in a server group for a different VRF, domain stripping is enabled in both the VRFs. VRF configurations are taken from server-group configuration

mode. If server-group configurations are disabled in global configuration mode but are available in server-group configuration mode, all configurations in server-group configuration mode are applicable.

After the domain stripping and broadcast accounting are configured, you can create separate accounting records as per the configurations.

If both **domain-stripping** and **directed-request** commands are enabled, domain stripping takes precedence and directed request functionality will not work.

How to Configure Authentication

Configuring Login Authentication Using AAA

The AAA security services facilitate a variety of login authentication methods. Use the **aaa authentication login** command to enable AAA authentication no matter which of the supported login authentication methods you decide to use. With the **aaa authentication login** command, you create one or more lists of authentication methods that are tried at login. These lists are applied using the **login authentication** line configuration command.

To configure login authentication by using AAA, use the following commands beginning in global configuration mode:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa new-model Example: Device(config)# aaa new-model	Enables AAA.
Step 4	aaa authentication login {default list-name} method1[method2...] Example: Device(config)# aaa authentication login default local	Creates a local authentication list.
Step 5	line [aux console tty vty] line-number [ending-line-number] Example: Device(config)# line vty 1	Enters line configuration mode for the lines to which you want to apply the authentication list.

	Command or Action	Purpose
Step 6	login authentication { default <i>list-name</i> } Example: Device(config-line)# login authentication default	Applies the authentication list to a line or set of lines.
Step 7	end Example: Device(config-line)# end	Exits line configuration mode and returns to privileged EXEC mode.

What to do next

The *list-name* is a character string used to name the list you are creating. The method argument refers to the actual method the authentication algorithm tries. The additional methods of authentication are used only if the previous method returns an error, not if it fails. To specify that the authentication should succeed even if all methods return an error, specify **none** as the final method in the command line.

For example, to specify that authentication should succeed even if (in this example) the TACACS+ server returns an error, enter the following command:

```
Device(config)# aaa authentication login default group tacacs+ none
```



Note Because the **none** keyword enables *any* user logging in to successfully authenticate, it should be used only as a backup method of authentication.

To create a default list that is used when a named list is *not* specified in the **login authentication** command, use the **default** keyword followed by the methods that are to be used in default situations. The default method list is automatically applied to all interfaces.

For example, to specify RADIUS as the default method for user authentication during login, enter the following command:

```
Device(config)# aaa authentication login default group radius
```

Configuring PPP Authentication Using AAA

The AAA security services facilitate a variety of authentication methods for use on serial interfaces running PPP. Use the **aaa authentication ppp** command to enable AAA authentication no matter which of the supported PPP authentication methods you decide to use.

To configure AAA authentication methods for serial lines using PPP, use the following commands in global configuration mode:

Procedure

	Command or Action	Purpose
Step 1	enable Example:	Enables privileged EXEC mode. Enter your password, if prompted.

	Command or Action	Purpose
	Device> enable	
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa new-model Example: Device(config)# aaa new-model	Enables AAA.
Step 4	aaa authentication ppp {default list-name} method1[method2...] Example: Device(config)# aaa authentication ppp-auth default local	Creates a local authentication list.
Step 5	interface interface-type interface-number Example: Device(config)# interface gigabitethernet 0/1/0	Enters interface configuration mode for the interface to which you want to apply the authentication list.
Step 6	ppp authentication {protocol1 [protocol2...]} [if-needed] {default list-name} [callin] [one-time][optional] Example: Device(config)# ppp authentication ms-chap ppp-auth	Applies the authentication list to a line or set of lines. In this command, <i>protocol1</i> and <i>protocol2</i> represent the following protocols: CHAP, MS-CHAP, and PAP. PPP authentication is attempted first using the first authentication method, specified by <i>protocol1</i> . If <i>protocol1</i> is unable to establish authentication, the next configured protocol is used to negotiate authentication.
Step 7	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

What to do next

With the **aaa authentication ppp** command, you create one or more lists of authentication methods that are tried when a user tries to authenticate via PPP. These lists are applied using the **ppp authentication** line configuration command.

To create a default list that is used when a named list is *not* specified in the **ppp authentication** command, use the **default** keyword followed by the methods you want used in default situations.

For example, to specify the local username database as the default method for user authentication, enter the following command:

```
Device(config)# aaa authentication ppp default local
```

The *list-name* is any character string used to name the list you are creating. The method argument refers to the actual method the authentication algorithm tries. The additional methods of authentication are used only if the previous method returns an error, not if it fails. To specify that the authentication should succeed even if all methods return an error, specify **none** as the final method in the command line.

For example, to specify that authentication should succeed even if (in this example) the TACACS+ server returns an error, enter the following command:

```
Device(config)# aaa authentication ppp default group tacacs+ none
```



Note Because **none** allows all users logging in to authenticate successfully, it should be used as a backup method of authentication.

Configuring ARAP Authentication Using AAA

Using the **aaa authentication arap** command, you can create one or more lists of authentication methods that are tried when AppleTalk Remote Access Protocol (ARAP) users attempt to log in to the device. These lists are used with the **arap authentication** line configuration command.

Use the following commands starting in global configuration mode:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa new-model Example: Device(config)# aaa new-model	Enables AAA.
Step 4	aaa authentication arap Example: Device(config)# aaa authentication arap Example: Enables authentication for ARAP users.	
Step 5	line number Example: Device(config)# line 1	(Optional) Changes to line configuration mode.

	Command or Action	Purpose
Step 6	autoselect arap Example: Device(config-line) # auto-select arap	(Optional) Enables autoselection of ARAP.
Step 7	autoselect during-login Example: Device(config-line) # autoselect during-login	(Optional) Starts the ARAP session automatically at user login.
Step 8	arap authentication list-name Example: Device(config-line) # arap authentication arap-authen	(Optional—not needed if default is used in the aaa authentication arap command) Enables TACACS+ authentication for ARAP on a line.
Step 9	end Example: Device(config-line) # end	Exits line configuration mode and returns to the privileged EXEC mode.

What to do next

The *list-name* is any character string used to name the list you are creating. The *method* argument refers to the actual list of methods the authentication algorithm tries, in the sequence entered.

To create a default list that is used when a named list is *not* specified in the **arap authentication** command, use the **default** keyword followed by the methods you want to use in default situations.

The additional methods of authentication are used only if the previous method returns an error, not if it fails. To specify that the authentication should succeed even if all methods return an error, specify **none** as the final method in the command line.



Note Because **none** allows all users logging in to be authenticated, it should be used as a backup method of authentication.

For example, to create a default AAA authentication method list used with ARAP, use the following command:

```
Device(config) # aaa authentication arap default if-needed none
```

To create the same authentication method list for ARAP and name the list *MIS-access*, use the following command:

```
Device(config) # aaa authentication arap MIS-access if-needed none
```

Configuring NASI Authentication Using AAA

Using the **aaa authentication nasi** command, you can create one or more lists of authentication methods that are tried when NetWare Asynchronous Services Interface (NASI) users attempt to log in to the device. These lists are used with the **nasi authentication line** configuration command.

To configure NASI authentication using AAA, use the following commands starting in global configuration mode:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa new-model Example: Device(config)# aaa new-model	Enables AAA globally.
Step 4	aaa authentication nasi Example: Device(config)# aaa authentication nasi	Enables authentication for NASI users.
Step 5	line number Example: Device(config)# line 4	(Optional--not needed if default is used in the aaa authentication nasi command.) Enters line configuration mode.
Step 6	nasi authentication list-name Example: Device(config-line)# nasi authentication nasi-authen	(Optional--not needed if default is used in the aaa authentication nasi command.) Enables authentication for NASI on a line.
Step 7	end Example: Device(config-line)# end	Exits line configuration mode and returns to the privileged EXEC mode.

What to do next

The *list-name* is any character string used to name the list you are creating. The *method* argument refers to the actual list of methods that the authentication algorithm tries, in the sequence entered.

To create a default list that is used when a named list is *not* specified in the **aaa authentication nasi** command, use the **default** keyword followed by the methods you want to use in default situations.

The additional methods of authentication are used only if the previous method returns an error, not if it fails. To specify that the authentication should succeed even if all methods return an error, specify **none** as the final method in the command line.



Note Because **none** allows all users logging in to be authenticated, it should be used as a backup method of authentication.

Preventing an Access Request with a Blank Username from Being Sent to the RADIUS Server

The following configuration steps provide the ability to prevent an Access Request with a blank username from being sent to the RADIUS server. This functionality ensures that unnecessary RADIUS server interaction is avoided, and RADIUS logs are kept short.

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa new-model Example: Device(config)# aaa new-model	Enables AAA globally.
Step 4	aaa authentication suppress null-username Example: Device(config)# aaa authentication suppress null-username	Prevents an Access Request with a blank username from being sent to the RADIUS server.
Step 5	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

Configuring Message Banners for AAA Authentication

AAA supports the use of configurable, personalized login and failed-login banners. You can configure message banners that will be displayed when a user logs in to the system to be authenticated using AAA and when, for whatever reason, authentication fails.

Configuring a Login Banner

To configure a banner that is displayed when a user logs in (replacing the default message for login), perform the following task:

Before you begin

To create a login banner, you must configure a delimiting character that notifies the system that the following text string must be displayed as the banner, and then the text string itself. The delimiting character is repeated at the end of the text string to signify the end of the banner. The delimiting character can be any single character in the extended ASCII character set, but once defined as the delimiter, that character cannot be used in the text string for the banner.

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa new-model Example: Device(config)# aaa new-model	Enables AAA.
Step 4	aaa authentication banner <i>delimiter string delimiter</i> Example: Device(config)# aaa authentication banner *Unauthorized use is prohibited.*	Creates a personalized login banner.
Step 5	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

Configuring a Failed-Login Banner

To configure a message that is displayed when a user login fails (replacing the default message for failed login), perform the following task:

Before you begin

To create a failed-login banner, you must configure a delimiting character, which notifies the system that the following text string must be displayed as the banner, and then configure the text string itself. The delimiting character is repeated at the end of the text string to signify the end of the failed-login banner. The delimiting

character can be any single character in the extended ASCII character set, but once defined as the delimiter, that character cannot be used in the text string making up the banner.

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa new-model Example: Device(config)# aaa new-model	Enables AAA.
Step 4	aaa authentication fail-message delimiter string delimiter Example: Device(config)# aaa authentication fail-message *Failed login. Try again.*	Creates a message to be displayed when a user login fails.
Step 5	end Example: Device(config)# end	Exits global configuration mode and returns to privileged EXEC mode.

Configuring AAA Packet of Disconnect

Packet of disconnect (POD) terminates connections on the network access server (NAS) when particular session attributes are identified. By using session information obtained from AAA, the POD client residing on a UNIX workstation sends disconnect packets to the POD server running on the network access server. The NAS terminates any inbound user session with one or more matching key attributes. It rejects requests when required fields are missing or when an exact match is not found.

To configure POD, perform the following tasks in global configuration mode:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.

	Command or Action	Purpose
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa accounting network default start-stop radius Example: Device(config)# aaa accounting network default start-stop radius	Enables AAA accounting records.
Step 4	aaa accounting delay-start Example: Device(config)# aaa accounting delay-start	(Optional) Delays generation of the start accounting record until the Framed-IP-Address is assigned, allowing its use in the POD packet.
Step 5	aaa pod server server-key string Example: Device(config)# aaa pod server server-key xyz123	Enables POD reception.
Step 6	radius server name non-standard Example: Device(config)# radius server radser	Configures a RADIUS server and enters RADIUS server configuration mode.
Step 7	address {ipv4 ipv6} hostname Example: Device(config-radius-server)# address ipv4 radius-host	Configures a RADIUS host.
Step 8	end Example: Device(config-radius-server)# end	Exits RADIUS server configuration mode and returns to privileged EXEC mode.

Configuring Double Authentication

To configure double authentication, perform the following steps:

1. Enable AAA by using the **aaa-new model** global configuration command.
2. Use the **aaa authentication** command to configure your network access server to use login and PPP authentication method lists, then apply those method lists to the appropriate lines or interfaces.
3. Use the **aaa authorization** command to configure AAA network authorization at login.
4. Configure security protocol parameters (for example, RADIUS or TACACS+).
5. Use access control list AV pairs on the security server that the user can connect to the local host only by establishing a Telnet connection.

6. (Optional) Configure the **access-profile** command as an autocommand. If you configure the autocommand, remote users will not have to manually enter the **access-profile** command to access authorized rights associated with their personal user profile.



Note If the **access-profile** command is configured as an autocommand, users will still have to Telnet to the local host and log in to complete double authentication.

Follow these rules when creating the user-specific authorization statements (These rules relate to the default behavior of the **access-profile** command):

- Use valid AV pairs when configuring access control list AV pairs on the security server.
- For remote users to use the interface's existing authorization (that which existed prior to the second stage authentication/authorization), but you want them to have different access control lists (ACLs), you should specify *only* ACL AV pairs in the user-specific authorization definition. This might be desirable if you set up a default authorization profile to apply to the remote host, but want to apply specific ACLs to specific users.
- When these user-specific authorization statements are later applied to the interface, they can either be *added* to the existing interface configuration or they can *replace* the existing interface configuration--depending on which form of the **access-profile** command is used to authorize the user. You should understand how the **access-profile** command works before configuring the authorization statements.
- If you will be using ISDN or Multilink PPP, you must also configure virtual templates at the local host.

Enabling Automated Double Authentication

You can make the double authentication process easier for users by implementing automated double authentication. Automated double authentication provides all of the security benefits of double authentication, but offers a simpler, more user-friendly interface for remote users. With double authentication, a second level of user authentication is achieved when the user Telnets to the network access server or router and enters a username and password. With automated double authentication, the user does not have to Telnet to the network access server; instead the user responds to a dialog box that requests a username and password or personal identification number (PIN). To use the automated double authentication feature, the remote user hosts must be running a companion client application.



Note Automated double authentication, like the existing double authentication feature, is for Multilink PPP ISDN connections only. Automated double authentication cannot be used with other protocols such as X.25 or SLIP.

Automated double authentication is an enhancement to the existing double authentication feature. To configure automated double authentication, you must first configure double authentication by completing the following steps:

1. Enable AAA by using the **aaa-new model** global configuration command.
2. Use the **aaa authentication** command to configure your network access server to use login and PPP authentication method lists, then apply those method lists to the appropriate lines or interfaces.
3. Use the **aaa authorization** command to configure AAA network authorization at login.

4. Configure security protocol parameters (for example, RADIUS or TACACS+).
5. Use access control list AV pairs on the security server that the user can connect to the local host only by establishing a Telnet connection.
6. Configure the **access-profile** command as an autocommand. If you configure the autocommand, remote users will not have to manually enter the **access-profile** command to access authorized rights associated with their personal user profile.



Note If the **access-profile** command is configured as an autocommand, users will still have to Telnet to the local host and log in to complete double authentication.

Follow these rules when creating the user-specific authorization statements (These rules relate to the default behavior of the **access-profile** command):

- Use valid AV pairs when configuring access control list AV pairs on the security server.
- If you want remote users to use the interface's existing authorization (that which existed prior to the second stage authentication/authorization), but you want them to have different access control lists (ACLs), you should specify *only* ACL AV pairs in the user-specific authorization definition. This might be desirable if you set up a default authorization profile to apply to the remote host, but want to apply specific ACLs to specific users.
- When these user-specific authorization statements are later applied to the interface, they can either be *added* to the existing interface configuration, or *replace* the existing interface configuration--depending on which form of the **access-profile** command is used to authorize the user. You should understand how the **access-profile** command works before configuring the authorization statements.
- If you will be using ISDN or Multilink PPP, you must also configure virtual templates at the local host.

Configuring Automated Double Authentication

To configure automated double authentication, perform the following task:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	ip trigger-authentication [timeout seconds] [port number] Example:	Enables automation of double authentication.

	Command or Action	Purpose
	Device(config)# ip trigger-authentication timeout 120	
Step 4	interface <i>type number</i> Example: Device(config)# interface gigabitethernet 1/0/17	Configures an interface and enter the interface configuration mode.
Step 5	ip trigger-authentication Example: Device(config-if)# ip trigger-authentication	Applies automated double authentication to the interface.
Step 6	end Example: Device(config-if)# end	Exits interface configuration mode and returns to privileged EXEC mode.

Troubleshooting Automated Double Authentication

To troubleshoot automated double authentication, use the following commands in privileged EXEC mode:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	show ip trigger-authentication Example: Device# show ip trigger-authentication	Displays the list of remote hosts for which automated double authentication has been attempted (successfully or unsuccessfully).
Step 3	clear ip trigger-authentication Example: Device# clear ip trigger-authentication	Clears the list of remote hosts for which automated double authentication has been attempted. (This clears the table displayed by the show ip trigger-authentication command.)
Step 4	debug ip trigger-authentication Example: Device# debug ip trigger-authentication	Displays debug output related to automated double authentication.

Configuring Domain Stripping at the Server Group Level

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	aaa group server radius server-name Example: Device(config)# aaa group server radius rad1	Adds the RADIUS server and enters server group RADIUS configuration mode. • The <i>server-name</i> argument specifies the RADIUS server group name.
Step 4	domain-stripping [strip-suffix word] [right-to-left] [prefix-delimiter word] [delimiter word] Example: Device(config-sg-radius)# domain-stripping delimiter username@example.com	Configures domain stripping at the server group level.
Step 5	end Example: Device(config-sg-radius)# end	Exits server group RADIUS configuration mode and returns to the privileged EXEC mode.

Configuring Non-AAA Authentication Methods

Configuring Line Password Protection

This task is used to provide access control on a terminal line by entering the password and establishing password checking.



Note

If you configure line password protection and then configure TACACS or extended TACACS, the TACACS username and password take precedence over line passwords. If you have not yet implemented a security policy, we recommend that you use AAA.

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	line [aux console tty vty] line-number [ending-line-number] Example: Device(config)# line console 0	Enters line configuration mode.
Step 4	password password Example: Device(config-line)# secret word	Assigns a password to a terminal or other device on a line. The password checker is case sensitive and can include spaces; for example, the password “Secret” is different from the password “secret,” and “two words” is an acceptable password.
Step 5	login Example: Device(config-line)# login	Enables password checking at login. You can disable line password verification by disabling password checking by using the no version of this command. Note The login command only changes username and privilege level but it does not execute a shell; therefore autocommands will not be executed. To execute autocommands under this circumstance, you need to establish a Telnet session back into the device (loop-back). Make sure that the device has been configured for secure Telnet sessions if you choose to implement autocommands this way.
Step 6	end Example: Device(config-line)# end	Exits line configuration mode and returns to privileged EXEC mode.

Establishing Username Authentication

You can create a username-based authentication system, which is useful in the following situations:

- To provide a TACACS-like username and encrypted password-authentication system for networks that cannot support TACACS
- To provide special-case logins: for example, access list verification, no password verification, autocommand execution at login, and “no escape” situations

To establish username authentication, perform the following task:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	Do one of the following: • username name [nopassword password password password encryption-type encrypted password] • username name [access-class number] Example: Device (config)# username superuser password superpassword password 7 encrypted-password Device (config)# username user1 access-class access-user	Establishes username authentication with encrypted passwords. or (Optional) Establishes username authentication by access list.
Step 4	username name [privilege level] Example: Device (config)# username user1 privilege 5	(Optional) Sets the privilege level for the user.
Step 5	username name [autocommand command] Example: Device (config)# username user1 autocommand show users	(Optional) Specifies a command to be executed automatically.
Step 6	username name [noescape] [nohangup] Example: Device (config)# username user1 noescape	(Optional) Sets a “no escape” login environment.
Step 7	end Example:	Exits global configuration mode and returns to privileged EXEC mode.

	Command or Action	Purpose
	Device(config)# end	

What to do next

The keyword **noescape** prevents users from using escape characters on the hosts to which they are connected. The **nohangup** feature does not disconnect after using the autocommand.



Caution Passwords will be displayed in clear text in your configuration unless you enable the **service password-encryption** command.

Defining PPP Authentication Using MS-CHAP

To define PPP authentication using MS-CHAP, use the following commands in interface configuration mode:

Procedure

	Command or Action	Purpose
Step 1	enable Example: Device> enable	Enables privileged EXEC mode. Enter your password, if prompted.
Step 2	configure terminal Example: Device# configure terminal	Enters global configuration mode.
Step 3	encapsulation ppp Example: Device(config)# encapsulation ppp	Enables PPP encapsulation.
Step 4	interface type number Example: Device(config)# interface gigabitethernet 1/0/2	Configures an interface and enters interface configuration mode.
Step 5	ppp authentication ms-chap [if-needed] [list-name default] [callin] [one-time] Example: Device(config-if)# ppp authentication ms-chap default callin	Defines PPP authentication using MS-CHAP.
Step 6	end Example: Device(config-if)# end	Exits interface configuration mode and returns to privileged EXEC mode.

What to do next

If you configure **ppp authentication ms-chap** on an interface, all incoming calls on that interface that initiate a PPP connection will have to be authenticated using MS-CHAP. If you configure the **ppp authentication** command with the **callin** keyword, the access server will only authenticate the remote device if the remote device initiated the call.

Authentication method lists and the **one-time** keyword are only available if you have enabled AAA—they will not be available if you are using TACACS or extended TACACS. If you specify the name of an authentication method list with the **ppp authentication** command, PPP will attempt to authenticate the connection using the methods defined in the specified method list. If AAA is enabled and no method list is defined by name, PPP will attempt to authenticate the connection using the methods defined as the default. The **ppp authentication** command with the **one-time** keyword enables support for one-time passwords during authentication.

The **if-needed** keyword is only available if you are using TACACS or extended TACACS. The **ppp authentication** command with the **if-needed** keyword means that PPP will only authenticate the remote device via MS-CHAP if that device has not yet authenticated during the life of the current call. If the remote device authenticated through a standard login procedure and initiated PPP from the EXEC prompt, PPP will not authenticate through MS-CHAP if **ppp authentication chap if-needed** is configured.



Note If PPP authentication using MS-CHAP is used with username authentication, you must include the MS-CHAP secret in the local username/password database.

Configuration Examples for Authentication

Example: Configuring Method Lists

Suppose the system administrator has decided on a security solution where all interfaces will use the same authentication methods to authenticate PPP connections. In the RADIUS group, R1 is contacted first for authentication information, then if there is no response, R2 is contacted. If R2 does not respond, T1 in the TACACS+ group is contacted; if T1 does not respond, T2 is contacted. If all designated servers fail to respond, authentication falls to the local username database on the access server itself. To implement this solution, the system administrator would create a default method list by entering the following command:

```
Device> enable
Device# configure terminal
Device(config)# aaa authentication ppp default group radius group tacacs+ local
Device(config)# exit
```

In this example, “default” is the name of the method list. The protocols included in this method list are listed after the name, in the order they are to be queried. The default list is automatically applied to all interfaces.

When a remote user attempts to dial in to the network, the network access server first queries R1 for authentication information. If R1 authenticates the user, it issues a PASS response to the network access server and the user is allowed to access the network. If R1 returns a FAIL response, the user is denied access and the session is terminated. If R1 does not respond, then the network access server processes that as an ERROR and queries R2 for authentication information. This pattern would continue through the remaining designated methods until the user is either authenticated or rejected, or until the session is terminated.

It is important to remember that a FAIL response is significantly different from an ERROR. A FAIL means that the user has not met the criteria contained in the applicable authentication database to be successfully authenticated. Authentication ends with a FAIL response. An ERROR means that the security server has not responded to an authentication query. Because of this, no authentication has been attempted. Only when an ERROR is detected will AAA select the next authentication method defined in the authentication method list.

Suppose the system administrator wants to apply a method list only to a particular interface or set of interfaces. In this case, the system administrator creates a named method list and then applies this named list to the applicable interfaces. The following example shows how the system administrator can implement an authentication method that will be applied only to interface 3:

```
Device> enable
Device# configure terminal
Device(config)# Device(config)#
Device(config)# aaa authentication ppp server-group1 group radius group tacacs+ local none
Device(config)# interface gigabitethernet 1/0/3
Device(config-if)# ppp authentication chap server-group1
Device(config-if)# end
```

In this example, “apple” is the name of the method list, and the protocols included in this method list are listed after the name in the order in which they are to be performed. After the method list has been created, it is applied to the appropriate interface. Note that the method list name (apple) in both the AAA and PPP authentication commands must match.

In the following example, the system administrator uses server groups to specify that only R2 and T2 are valid servers for PPP authentication. To do this, the administrator must define specific server groups whose members are R2 (172.16.2.7) and T2 (172.16.2.77), respectively. In this example, the RADIUS server group “rad2only” is defined as follows using the **aaa group server** command:

```
Device> enable
Device# configure terminal
Device(config)# aaa group server radius rad2only
Device(config-sg-radius)# server 172.16.2.7
Device(config-sg-radius)# end
```

The TACACS+ server group “tac2only” is defined as follows using the **aaa group server** command:

```
Device> enable
Device# configure terminal
Device(config)# aaa group server tacacs+ tac2only
Device(config-sg-tacacs)# server 172.16.2.77
Device(config-sg-tacacs)# end
```

The administrator then applies PPP authentication using the server groups. In this example, the default methods list for PPP authentication follows this order: **group rad2only**, **group tac2only**, and **local**:

```
Device> enable
Device# configure terminal
Device(config)# aaa authentication ppp default group rad2only group tac2only local
Device(config)# exit
```

If a method list is configured under VTY lines, the corresponding method list must be added to AAA. The following example shows how to configure a method list under a VTY line:

```
Device> enable
Device# configure terminal
Device(config)# line vty 0 4
Device(config-line)# authorization commands 15 auth1
Device(config-line)# exit
```

The following example shows how to configure a method list in AAA:

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# aaa authorization commands 15 auth1 group tacacs+
Device(config)# exit
```

If no method list is configured under VTY lines, the default method list must be added to AAA. The following example shows a VTY configuration without a method list:

```
Device> enable
Device# configure terminal
Device(config)# line vty 0 4
Device(config-line)# end
```

The following example shows how to configure the default method list:

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# aaa authorization commands 15 default group tacacs+
Device(config)# exit
```

Example: RADIUS Authentication

This section provides two sample configurations using RADIUS.

The following example shows how to configure the router to authenticate and authorize using RADIUS:

```
Device> enable
Device# configure terminal
Device(config)# aaa authentication login radius-login group radius local
Device(config)# aaa authentication ppp radius-ppp if-needed group radius
Device(config)# aaa authorization exec default group radius if-authenticated
Device(config)# aaa authorization network default group radius
Device(config)# line 3
Device(config-line)# login authentication radius-login
Device(config-line)# exit
Device(config)# interface gigabitethernet 1/0/1
Device(config-if)# ppp authentication radius-ppp
Device(config-if)# end
```

The lines in this sample RADIUS authentication and authorization configuration are defined as follows:

- The **aaa authentication login radius-login group radius local** command configures the router to use RADIUS for authentication at the login prompt. If RADIUS returns an error, the user is authenticated using the local database.
- The **aaa authentication ppp radius-ppp if-needed group radius** command configures the Cisco IOS XE software to use PPP authentication using CHAP or PAP if the user has not already logged in. If the EXEC facility has authenticated the user, PPP authentication is not performed.
- The **aaa authorization exec default group radius if-authenticated** command queries the RADIUS database for information that is used during EXEC authorization, such as autocommands and privilege levels, but only provides authorization if the user has successfully authenticated.
- The **aaa authorization network default group radius** command queries RADIUS for network authorization, address assignment, and other access lists.
- The **login authentication radius-login** command enables the radius-login method list for line 3.
- The **ppp authentication radius-ppp** command enables the radius-ppp method list for serial interface 0.

The following example shows how to configure the router to prompt for and verify a username and password, authorize the user's EXEC level, and specify it as the method of authorization for privilege level 2. In this example, if a local username is entered at the username prompt, that username is used for authentication.

If the user is authenticated using the local database, EXEC authorization using RADIUS will fail because no data is saved from the RADIUS authentication. The method list also uses the local database to find an autocommand. If there is no autocommand, the user becomes the EXEC user. If the user then attempts to issue commands that are set at privilege level 2, TACACS+ is used to attempt to authorize the command.

```
Device> enable
Device# configure terminal
Device(config)# aaa authentication login default group radius local
Device(config)# aaa authorization exec default group radius local
Device(config)# aaa authorization command 2 default group tacacs+ if-authenticated
Device(config)# radius server radserver
Device(config-sg-radius)# address ipv4 10.2.3.1
Device(config-sg-radius)# exit
Device(config)# radius-server attribute 44 include-in-access-req
Device(config)# radius-server attribute 8 include-in-access-req
Device(config)# end
```

The lines in this sample RADIUS authentication and authorization configuration are defined as follows:

- The `aaa authentication login default group radius local` command specifies that the username and password are verified by RADIUS or, if RADIUS is not responding, by the router's local user database.
- The `aaa authorization exec default group radius local` command specifies that RADIUS authentication information be used to set the user's EXEC level if the user authenticates with RADIUS. If no RADIUS information is used, this command specifies that the local user database be used for EXEC authorization.
- The `aaa authorization command 2 default group tacacs+ if-authenticated` command specifies TACACS+ authorization for commands set at privilege level 2, if the user has already successfully authenticated.
- The `radius-server attribute 44 include-in-access-req` command sends RADIUS attribute 44 (Acct-Session-ID) in access-request packets.
- The `radius-server attribute 8 include-in-access-req` command sends RADIUS attribute 8 (Framed-IP-Address) in access-request packets.

Example: TACACS Authentication

The following example shows how to configure TACACS+ as the security protocol to be used for PPP authentication:

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# aaa authentication ppp test group tacacs+ local
Device(config)# interface gigabitethernet 1/1/2
Device(config-if)# ppp authentication chap pap test
Device(config-if)# exit
Device(config)# tacacs server server1
Device(config-server-tacacs)# address ipv4 192.0.2.3
Device(config-server-tacacs)# key key1
Device(config-server-tacacs)# end
```

The lines in this sample TACACS+ authentication configuration are defined as follows:

- The `aaa new-model` command enables the AAA security services.

- The **aaa authentication** command defines a method list, “test,” to be used on serial interfaces running PPP. The keywords **group tacacs+** means that authentication will be done through TACACS+. If TACACS+ returns an ERROR of some sort during authentication, the keyword **local** indicates that authentication will be attempted using the local database on the network access server.
- The **interface** command selects the line.
- The **ppp authentication** command applies the test method list to this line.
- The **address ipv4** command identifies the TACACS+ daemon as having an IP address of 192.0.2.3.
- The **key** command defines the shared encryption key to be “key1.”

The following example shows how to configure AAA authentication for PPP:

```
Device(config)# aaa authentication ppp default if-needed group tacacs+ local
```

In this example, the keyword **default** means that PPP authentication is applied by default to all interfaces. The **if-needed** keyword means that if the user has already authenticated by going through the ASCII login procedure, then PPP is not necessary and can be skipped. If authentication is needed, the keywords **group tacacs+** means that authentication will be done through TACACS+. If TACACS+ returns an ERROR of some sort during authentication, the keyword **local** indicates that authentication will be attempted using the local database on the network access server.

The following example shows how to create the same authentication algorithm for PAP, but it calls the method list “MIS-access” instead of “default”:

```
Device> enable
Device# configure terminal
Device(config)# aaa authentication ppp MIS-access if-needed group tacacs+ local
Device(config)# interface gigabitethernet 1/1/2
Device(config)# ppp authentication pap MIS-access
Device(config)# end
```

In this example, because the list does not apply to any interfaces (unlike the default list, which applies automatically to all interfaces), the administrator must select interfaces to which this authentication scheme should apply by using the **interface** command. The administrator must then apply this method list to those interfaces by using the **ppp authentication** command.

Example: AAA Scalability

The following example shows a general security configuration using AAA with RADIUS as the security protocol. In this example, the network access server is configured to allocate 16 background processes to handle AAA requests for PPP.

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# radius server radserver
Device(config-sg-radius)# address ipv4 radius-host
Device(config-sg-radius)# key myRaDiUSpassWoRd
Device(config-sg-radius)# exit
Device(config)# radius-server configure-nas
Device(config)# username root password ALongPassword
Device(config)# aaa authentication ppp dialins group radius local
Device(config)# aaa authentication login admins local
Device(config)# aaa authorization network default group radius local
Device(config)# aaa accounting network default start-stop group radius
Device(config)# aaa processes 16
```

```

Device(config)# line 1 16
Device(config-line)# autoselect ppp
Device(config-line)# autoselect during-login
Device(config-line)# login authentication admins
Device(config-line)# modem dialin
Device(config-line)# exit
Device(config)# interface gigabitethernet 1/2/0
Device(config-if)# group-range 1 16
Device(config-if)# encapsulation ppp
Device(config-if)# ppp authentication pap dialins
Device(config-if)# end

```

The lines in this sample RADIUS AAA configuration are defined as follows:

- The **aaa new-model** command enables AAA network security services.
- The **address ipv4 {hostname | host-address}** command defines the name of the RADIUS server host.
- The **key** command defines the shared secret text string between the network access server and the RADIUS server host.
- The **radius-server configure-nas** command defines that the Cisco router or access server will query the RADIUS server for static routes and IP pool definitions when the device first starts up.
- The **username** command defines the username and password to be used for the PPP Password Authentication Protocol (PAP) caller identification.
- The **aaa authentication ppp dialins group radius local** command defines the authentication method list “dialins,” which specifies that RADIUS authentication, then (if the RADIUS server does not respond) local authentication will be used on serial lines using PPP.
- The **aaa authentication login admins local** command defines another method list, “admins,” for login authentication.
- The **aaa authorization network default group radius local** command is used to assign an address and other network parameters to the RADIUS user.
- The **aaa accounting network default start-stop group radius** command tracks PPP usage.
- The **aaa processes** command allocates 16 background processes to handle AAA requests for PPP.
- The **line** command switches the configuration mode from global configuration to line configuration and identifies the specific lines being configured.
- The **autoselect ppp** command allows a PPP session to start up automatically on these selected lines.
- The **autoselect during-login** command is used to display the username and password prompt without pressing the Return key. After the user logs in, the autoselect function (in this case, PPP) begins.
- The **login authentication admins** command applies the “admins” method list for login authentication.
- The **modem dialin** command configures modems attached to the selected lines to only accept incoming calls.
- The **interface group-async** command selects and defines an asynchronous interface group.
- The **group-range** command defines the member asynchronous interfaces in the interface group.
- The **encapsulation ppp** command sets PPP as the encapsulation method used on the specified interfaces.
- The **ppp authentication pap dialins** command applies the “dialins” method list to the specified interfaces.

Example: Configuring Login and Failed-Login Banners for AAA Authentication

The following example shows how to configure a login banner that is displayed when a user logs in to the system, (in this case, the phrase “Unauthorized Access Prohibited”). The asterisk (*) is used as the delimiting character. RADIUS is specified as the default login authentication method.

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# aaa authentication banner *Unauthorized Access Prohibited*
Device(config)# aaa authentication login default group radius
Device(config)# end
```

This configuration displays the following login banner:

```
Unauthorized Access Prohibited
Username:
```

The following example shows how to configure a failed-login banner that is displayed when a user tries to log in to the system and fails, (in this case, the phrase “Failed login. Try again”). The asterisk (*) is used as the delimiting character. RADIUS is specified as the default login authentication method.

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# aaa authentication banner *Unauthorized Access Prohibited*
Device(config)# aaa authentication fail-message *Failed login. Try again.*
Device(config)# aaa authentication login default group radius
Device(config)# end
```

This configuration displays the following login and failed-login banner:

```
Unauthorized Access Prohibited
Username:
Password:
Failed login. Try again.
```

Example: AAA Packet of Disconnect Server Key

The following example shows how to configure POD (packet of disconnect), which terminates connections on the network access server (NAS) when particular session attributes are identified.

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# aaa authentication ppp default radius
Device(config)# aaa accounting network default start-stop radius
Device(config)# aaa accounting delay-start
Device(config)# aaa pod server server-key xyz123
Device(config)# radius server non-standard
Device(config-sg-radius)# address ipv4 10.2.1.1
Device(config-sg-radius)# key rad123
Device(config-sg-radius)# end
```

Example: Double Authentication

The examples in this section illustrate possible configurations to be used with double authentication. Your configurations could differ significantly, depending on your network and security requirements.



Note These configuration examples include specific IP addresses and other specific information. This information is for illustration purposes only: your configuration will use different IP addresses, different usernames and passwords, and different authorization statements.

Example: Configuration of the Local Host for AAA with Double Authentication

These two examples show how to configure a local host to use AAA for PPP and login authentication, and for network and EXEC authorization. An example each is shown for RADIUS and for TACACS+.

In both the examples, the first three lines configure AAA with a specific server as the AAA server. The next two lines configure AAA for PPP and login authentication, and the last two lines configure network and EXEC authorization. The last line is necessary only if the **access-profile** command will be executed as an autocommand.

The following example shows device configuration with a RADIUS AAA server:

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# radius server radserver
Device(config-sg-radius)# address ipv4 secureserver
Device(config-sg-radius)# key myradiuskey
Device(config-sg-radius)# exit
Device(config)# aaa authentication ppp default group radius
Device(config)# aaa authentication login default group radius
Device(config)# aaa authorization network default group radius
Device(config)# aaa authorization exec default group radius
Device(config)# end
```

The following example shows device configuration with a TACACS+ server:

```
Device> enable
Device# configure terminal
Device(config)# aaa new-model
Device(config)# tacacs server server1
Device(config-server-tacacs)# address ipv4 192.0.2.3
Device(config-server-tacacs)# key mytacacskey
Device(config-server-tacacs)# exit
Device(config)# aaa authentication ppp default group tacacs+
Device(config)# aaa authentication login default group tacacs+
Device(config)# aaa authorization network default group tacacs+
Device(config)# aaa authorization exec default group tacacs+
Device(config)# end
```

Example: Configuration of the AAA Server for First-Stage PPP Authentication and Authorization

This example shows a configuration on the AAA server. A partial sample AAA configuration is shown for RADIUS.

TACACS+ servers can be configured similarly. (See the Complete Configuration with TACACS Example.)

This example defines authentication/authorization for a remote host named “hostx” that will be authenticated by CHAP in the first stage of double authentication. Note that the ACL AV pair limits the remote host to Telnet connections to the local host. The local host has the IP address 10.0.0.2.

The following example shows a partial AAA server configuration for RADIUS:

```

hostx Password = "welcome"
      User-Service-Type = Framed-User,
      Framed-Protocol = PPP,
      cisco-avpair = "lcp:interface-config=ip unnumbered fastethernet 0",
      cisco-avpair = "ip:inacl#3=permit tcp any 172.21.114.0 0.0.0.255 eq telnet",
      cisco-avpair = "ip:inacl#4=deny icmp any any",
      cisco-avpair = "ip:route#5=10.0.0.0 255.0.0.0",
      cisco-avpair = "ip:route#6=10.10.0.0 255.0.0.0",
      cisco-avpair = "ipx:inacl#3=deny any"

```

Example: Configuration of the AAA Server for Second-Stage Per-User Authentication and Authorization

This section contains partial sample AAA configurations on a RADIUS server. These configurations define authentication and authorization for a user with the username "user1," who will be user-authenticated in the second stage of double authentication.

TACACS+ servers can be configured similarly.

Three examples show sample RADIUS AAA configurations that could be used with each of the three forms of the **access-profile** command.

The first example shows a partial sample AAA configuration that works with the default form (no keywords) of the **access-profile** command. Note that only ACL AV pairs are defined. This example also sets up the **access-profile** command as an autocommand.

```

user1 Password = "welcome"
      User-Service-Type = Shell-User,
      cisco-avpair = "shell:autocmd=access-profile"
      User-Service-Type = Framed-User,
      Framed-Protocol = PPP,
      cisco-avpair = "ip:inacl#3=permit tcp any host 10.0.0.2 eq telnet",
      cisco-avpair = "ip:inacl#4=deny icmp any any"

```

The second example shows a partial sample AAA configuration that works with the **access-profile merge** form of the **access-profile** command. This example also sets up the **access-profile merge** command as an autocommand.

```

user1 Password = "welcome"
      User-Service-Type = Shell-User,
      cisco-avpair = "shell:autocmd=access-profile merge"
      User-Service-Type = Framed-User,
      Framed-Protocol = PPP,
      cisco-avpair = "ip:inacl#3=permit tcp any any",
      cisco-avpair = "ip:route=10.0.0.0 255.255.0.0",
      cisco-avpair = "ip:route=10.1.0.0 255.255.0.0",
      cisco-avpair = "ip:route=10.2.0.0 255.255.0.0"

```

The third example shows a partial sample AAA configuration that works with the **access-profile replace** form of the **access-profile** command. This example also sets up the **access-profile replace** command as an autocommand.

```

user1 Password = "welcome"
      User-Service-Type = Shell-User,
      cisco-avpair = "shell:autocmd=access-profile replace"
      User-Service-Type = Framed-User,
      Framed-Protocol = PPP,
      cisco-avpair = "ip:inacl#3=permit tcp any any",
      cisco-avpair = "ip:inacl#4=permit icmp any any",
      cisco-avpair = "ip:route=10.10.0.0 255.255.0.0",
      cisco-avpair = "ip:route=10.11.0.0 255.255.0.0",
      cisco-avpair = "ip:route=10.12.0.0 255.255.0.0"

```

Example: Complete Configuration with TACACS

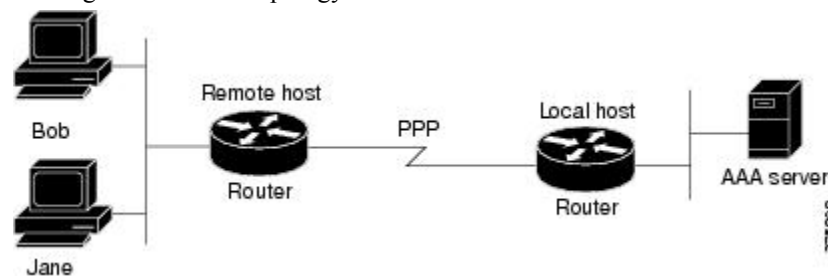
This example shows TACACS+ authorization profile configurations both for the remote host (used in the first stage of double authentication) and for specific users (used in the second stage of double authentication).

This sample configuration shows authentication/authorization profiles on the TACACS+ server for the remote host “hostx” and for three users, with the usernames “user_default,” “user_merge,” and “user_replace.” The configurations for these three usernames illustrate different configurations that correspond to the three different forms of the **access-profile** command. The three user configurations also illustrate setting up the autocommand for each form of the **access-profile** command.

The figure below shows the topology. The example that follows the figure shows a TACACS+ configuration file.

Figure 3: Topology Example for Double Authentication

This figure shows the topology of double authentication.



This sample configuration shows authentication/authorization profiles on the TACACS+ server for the remote host “hostx” and for three users, with the usernames “user_default,” “user_merge,” and “user_replace.”

```

key = "mytacacskey"
default authorization = permit
#-----Remote Host (BRI)-----
#
# This allows the remote host to be authenticated by the local host
# during fist-stage authentication, and provides the remote host
# authorization profile.
#
#-----
user = hostx
{
    login = cleartext "welcome"
    chap = cleartext "welcome"
    service = ppp protocol = lcp {
        interface-config="ip unnumbered fastethernet 0"
    }
    service = ppp protocol = ip {
        # It is important to have the hash sign and some string after
        # it. This indicates to the NAS that you have a per-user
        # config.
        inacl#3="permit tcp any 172.21.114.0 0.0.0.255 eq telnet"
        inacl#4="deny icmp any any"
        route#5="10.0.0.0 255.0.0.0"
        route#6="10.10.0.0 255.0.0.0"
    }
    service = ppp protocol = ipx {
        # see previous comment about the hash sign and string, in protocol = ip
        inacl#3="deny any"
    }
}
#----- "access-profile" default user "only acIs" -----

```

Example: Complete Configuration with TACACS

```

#
# Without arguments, access-profile removes any access-lists it can find
# in the old configuration (both per-user and per-interface), and makes sure
# that the new profile contains ONLY access-list definitions.
#
#-----
user = user_default
{
    login = cleartext "welcome"
    chap = cleartext "welcome"
    service = exec
    {
        # This is the autocommand that executes when user_default logs in.
        autocmd = "access-profile"
    }
    service = ppp protocol = ip {
        # Put whatever access-lists, static routes, whatever
        # here.
        # If you leave this blank, the user will have NO IP
        # access-lists (not even the ones installed prior to
        # this)!
        inacl#3="permit tcp any host 10.0.0.2 eq telnet"
        inacl#4="deny icmp any any"
    }
    service = ppp protocol = ipx {
        # Put whatever access-lists, static routes, whatever
        # here.
        # If you leave this blank, the user will have NO IPX
        # access-lists (not even the ones installed prior to
        # this)!
    }
}
#----- "access-profile merge" user -----
#
# With the 'merge' option, first all old access-lists are removed (as before),
# but then (almost) all AV pairs are uploaded and installed. This will allow
# for uploading any custom static routes, sap-filters, and so on, that the user
# may need in his or her profile. This needs to be used with care, as it leaves
# open the possibility of conflicting configurations.
#
#-----
user = user_merge
{
    login = cleartext "welcome"
    chap = cleartext "welcome"
    service = exec
    {
        # This is the autocommand that executes when user_merge logs in.
        autocmd = "access-profile merge"
    }
    service = ppp protocol = ip
    {
        # Put whatever access-lists, static routes, whatever
        # here.
        # If you leave this blank, the user will have NO IP
        # access-lists (not even the ones installed prior to
        # this)!
        inacl#3="permit tcp any any"
        route#2="10.0.0.0 255.255.0.0"
        route#3="10.1.0.0 255.255.0.0"
        route#4="10.2.0.0 255.255.0.0"
    }
    service = ppp protocol = ipx
    {

```

```

        # Put whatever access-lists, static routes, whatever
        # here.
        # If you leave this blank, the user will have NO IPX
        # access-lists (not even the ones installed prior to
        # this)!
    }
}
#----- "access-profile replace" user -----
#
# With the 'replace' option, ALL old configuration is removed and ALL new
# configuration is installed.
#
# One caveat: access-profile checks the new configuration for address-pool and
# address AV pairs. As addresses cannot be renegotiated at this point, the
# command will fail (and complain) when it encounters such an AV pair.
# Such AV pairs are considered to be "invalid" for this context.
#-----
user = user_replace
{
    login = cleartex
t
"
welcome
"

    chap = cleartext "welcome"
    service = exec
    {
        # This is the autocommand that executes when user_replace logs in.
        autocmd = "access-profile replace"
    }
    service = ppp protocol = ip
    {
        # Put whatever access-lists, static routes, whatever
        # here.
        # If you leave this blank, the user will have NO IP
        # access-lists (not even the ones installed prior to
        # this)!
        inacl#3="permit tcp any any"
        inacl#4="permit icmp any any"
        route#2="10.10.0.0 255.255.0.0"
        route#3="10.11.0.0 255.255.0.0"
        route#4="10.12.0.0 255.255.0.0"
    }
    service = ppp protocol = ipx
    {
        # put whatever access-lists, static routes, whatever
        # here.
        # If you leave this blank, the user will have NO IPX
        # access-lists (not even the ones installed prior to
        # this)!
    }
}
}

```

Example: Automated Double Authentication

This example shows a complete configuration file with automated double authentication configured. The configuration commands that apply to automated double authentication are preceded by descriptions with a double asterisk (**).

Current configuration:
!

Example: Automated Double Authentication

```

version 16.10
no service password-encryption
!
hostname myrouter
!
!
! **The following AAA commands are used to configure double authentication:
!
! **The following command enables AAA:
aaa new-model
! **The following command enables user authentication via the RADIUS AAA server:
!
aaa authentication login default none
aaa authentication ppp default group radius
! **The following command causes the remote user's authorization profile to be
! downloaded from the AAA server to the router when required:
!
aaa authorization network default group radius
!
enable password mypassword
!
ip host blue 172.21.127.226
ip host green 172.21.127.218
ip host red 172.21.127.114
ip domain-name example.com
ip name-server 172.16.2.75
!
!
interface GigabitEthernet0/0/0
ip address 172.21.127.186 255.255.255.248
no ip route-cache
no ip mroute-cache
no keepalive
ntp disable
no cdp enable
!
interface Virtual-Template1
ip unnumbered loopback0
no ip route-cache
no ip mroute-cache
!
! **The following command specifies that device authentication occurs via PPP CHAP:
ppp authentication chap
!
router eigrp 109
network 172.21.0.0
no auto-summary
!
ip default-gateway 172.21.127.185
no ip classless
ip route 172.21.127.114 255.255.255.255 172.21.127.113
! **Virtual profiles are required for double authentication to work:
virtual-profile virtual-template 1
dialer-list 1 protocol ip permit
no cdp run
! **The following command defines where the TACACS+ AAA server is:
tacacs server server1
address ipv4 172.16.57.35
! **The following command defines the key to use with TACACS+ traffic (required):
key mytacacskey
snmp-server community public RO
!
line con 0
exec-timeout 0 0

```

```
login authentication console
line aux 0
  transport input all
line vty 0 4
  exec-timeout 0 0
  password lab
!
end
```

Feature History for Configuring Authentication

This table provides release and related information for the features explained in this module.

These features are available in all the releases subsequent to the one they were introduced in, unless noted otherwise.

Release	Feature	Feature Information
Cisco IOS XE Gibraltar 16.11.1	AAA Authentication	Authentication provides a method to identify users, which includes the login and password dialog, challenge and response, messaging support, and encryption, depending on the selected security protocol. Authentication is the way a user is identified prior to being allowed access to the network and network services.
Cisco IOS XE Cupertino 17.7.1	AAA Authentication	This feature was implemented on the Cisco Catalyst 9600 Series Supervisor 2 Module (C9600X-SUP-2).

Use the Cisco Feature Navigator to find information about platform and software image support. To access Cisco Feature Navigator, go to <https://cfng.cisco.com/>.

